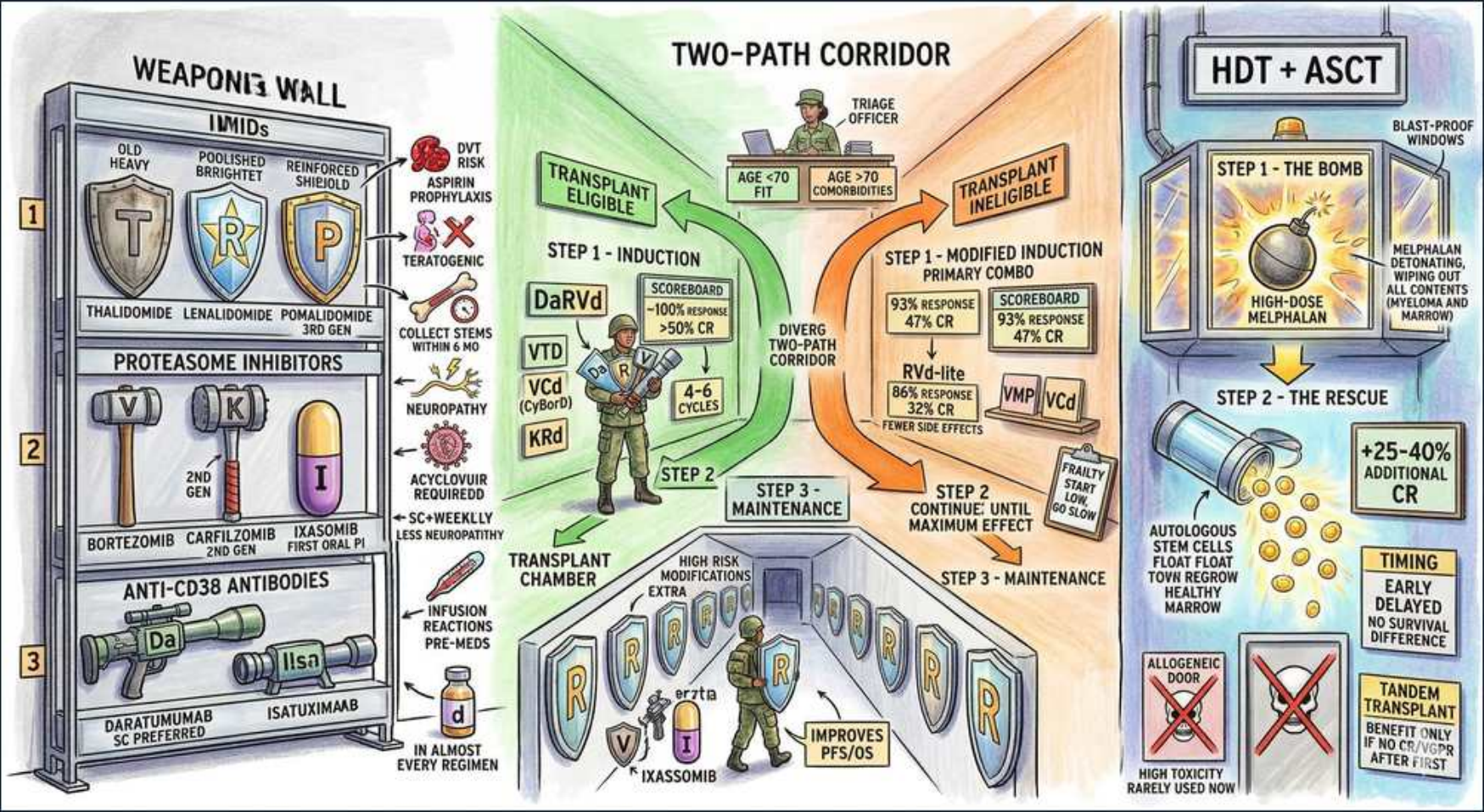


Plasma Cell — Treatment Backbone & Transplant



1. IMiDs class

WHAT YOU SEE

Weapons-wall shields T/R/P (thalidomide, lenalidomide, pomalidomide)

WHAT IT ENCODES

Immunomodulatory drugs (IMiDs) — Thalidomide, lenalidomide (Revlimid), Pomalidomide — are a core myeloma backbone. They bind CEREBLON, redirecting the E3 ubiquitin ligase to degrade transcription factors IKZF1/IKZF3 (Ikaros/Aiolos), killing plasma cells AND stimulating T/NK cells. Lenalidomide is the workhorse, especially for MAINTENANCE. Mnemonic: T-R-P shields = the three IMiDs; cereblon is the lock they fit.

BOARD TRAP

WRONG answer: 'IMiDs and proteasome inhibitors are interchangeable / share the same mechanism, so failing one predicts failing the other.' Different mechanisms (cereblon/IKZF degradation vs. proteasome blockade) — that's exactly why frontline regimens COMBINE them (e.g., VRd). Discriminator: lenalidomide-refractory disease still responds to a proteasome inhibitor or pomalidomide; do not assume cross-resistance.

2. IMiD = VTE risk

WHAT YOU SEE

DVT-risk skull + aspirin prophylaxis label

WHAT IT ENCODES

All IMiDs (thalidomide, lenalidomide, pomalidomide) substantially raise VENOUS THROMBOEMBOLISM risk, amplified when combined with high-dose dexamethasone or erythropoiesis-stimulating agents. Every patient needs thromboprophylaxis: aspirin for low-risk patients, and LMWH or a DOAC (e.g., apixaban) for high-risk patients (prior VTE, immobility, high-risk regimen). Mnemonic: the DVT skull warns you to add prophylaxis before starting an IMiD.

BOARD TRAP

WRONG answer: 'IMiDs cause cytopenias and bleeding, so anticoagulation/aspirin is contraindicated.' The dominant vascular hazard is THROMBOSIS, not bleeding — omitting prophylaxis is the classic error. Discriminator: risk-stratify and choose ASA (low risk) vs LMWH/DOAC (high risk); thrombocytopenia <50k may modify but does not eliminate the need to consider prophylaxis.

3. Thalidomide teratogen

WHAT YOU SEE

Teratogenic warning by thalidomide

WHAT IT ENCODES

Thalidomide is the infamous teratogen causing phocomelia (limb-shortening); lenalidomide and pomalidomide are likewise strictly teratogenic. All are dispensed under a mandatory REMS pregnancy-prevention program requiring two reliable contraception methods, regular pregnancy testing, and counseling. Thalidomide additionally causes sedation, constipation, and peripheral neuropathy. Mnemonic: the teratogen warning sign is nailed right next to thalidomide.

BOARD TRAP

WRONG answer: 'only thalidomide is teratogenic; lenalidomide and pomalidomide are safe in pregnancy.' ALL IMiDs are teratogenic and all require the REMS/pregnancy-prevention program. Discriminator: never start any IMiD in a person of childbearing potential without documented contraception and negative pregnancy testing.

4. Lenalidomide neuropathy-sparing

WHAT YOU SEE

Neuropathy note on IMiD wall

WHAT IT ENCODES

Lenalidomide causes LITTLE peripheral neuropathy (unlike thalidomide and the PI bortezomib) — its main toxicities are MYELOSUPPRESSION (neutropenia, thrombocytopenia), VTE, diarrhea, rash, and a small increased risk of second primary malignancies with prolonged use. This favorable neuro profile makes it ideal for long-term maintenance. Mnemonic: lenalidomide spares the nerves but hits the marrow.

BOARD TRAP

WRONG answer: 'new peripheral neuropathy in a patient on lenalidomide maintenance is from the lenalidomide.' Lenalidomide rarely causes neuropathy — suspect THALIDOMIDE or BORTEZOMIB instead, or another cause. Discriminator: lenalidomide's signature problems are cytopenias and VTE; stocking-glove neuropathy points to thalidomide/bortezomib.

5. Proteasome inhibitors

WHAT YOU SEE

Hammer rack — bortezomib, carfilzomib, ixazomib

WHAT IT ENCODES

Proteasome inhibitors (PIs) — Bortezomib (reversible, IV/SC), Carfilzomib (irreversible, IV), Ixazomib (reversible, ORAL) — block the 26S proteasome's chymotryptic activity, so misfolded/ubiquitinated proteins accumulate and trigger plasma-cell apoptosis (plasma cells are exquisitely sensitive due to heavy immunoglobulin synthesis). Bortezomib is the prototype INDUCTION drug. Mnemonic: the hammer rack smashes the proteasome 'garbage disposal.'

BOARD TRAP

WRONG answer: 'all proteasome inhibitors share the same toxicity profile.' They differ critically: bortezomib → peripheral neuropathy; carfilzomib → cardiotoxicity (sparing nerves); ixazomib → GI/thrombocytopenia. Discriminator: match the adverse effect to the agent — neuropathy = bortezomib, heart failure/HTN = carfilzomib, oral convenience = ixazomib.

6. Bortezomib neuropathy

WHAT YOU SEE

SC-weekly / neuropathy note near bortezomib

WHAT IT ENCODES

Bortezomib's dose-limiting toxicity is painful PERIPHERAL NEUROPATHY; giving it SUBCUTANEOUS and WEEKLY (rather than IV twice-weekly) markedly reduces neuropathy without losing efficacy. It also reactivates VARICELLA-ZOSTER, so antiviral prophylaxis (acyclovir/valacyclovir) is required. Mnemonic: SC-weekly bortezomib + acyclovir shield = fewer nerves damaged, no shingles.

BOARD TRAP

WRONG answer: 'a patient starting bortezomib doesn't need antiviral prophylaxis.' PIs (especially bortezomib) markedly raise herpes-zoster reactivation risk — acyclovir/valacyclovir prophylaxis is standard. Discriminator: bortezomib = neuropathy + zoster risk (give SC weekly + acyclovir); carfilzomib substitutes cardiotoxicity for the neuropathy.

7. Anti-CD38

WHAT YOU SEE

Daratumumab / isatuximab rockets

WHAT IT ENCODES

Anti-CD38 monoclonal antibodies — daratumumab (SC preferred over IV) and isatuximab — target CD38 on plasma cells (ADCC/ADCP/CDC/direct apoptosis) and are now in NEARLY EVERY regimen, including frontline QUADRUPLLET therapy (Dara-VRd / Dara-RVd). Premedicate to limit infusion reactions; expect neutropenia and infection risk. Mnemonic: CD38 rockets are now part of almost every launch.

BOARD TRAP

WRONG answer: 'an unexpected positive antibody screen / interference on the M-protein assay after starting daratumumab is a transfusion alloantibody or treatment failure.' Daratumumab binds CD38 on reagent RBCs → FALSE-positive indirect Coombs/antibody screen (mitigate with DTT-treated cells), and it migrates as IgG-kappa on immunofixation, mimicking/masking the M-spike. Discriminator: phenotype/genotype the patient BEFORE starting; use daratumumab-specific assays (e.g., DIRA) to interpret the M-protein.

8. Transplant-eligible induction

WHAT YOU SEE

Transplant-eligible quad regimens Dara-RVd / VRd / KRd

WHAT IT ENCODES

Fit, transplant-ELIGIBLE patients get a 3- or 4-drug INDUCTION combining a PI + IMiD + steroid ± anti-CD38 — e.g., VRd (bortezomib-lenalidomide-dexamethasone) or quadruplet Dara-VRd, sometimes KRd — for roughly 4 cycles, then stem-cell collection, then autologous transplant. Combining classes attacks via complementary mechanisms for deeper response. Mnemonic: the transplant-eligible booth stocks VRd/Dara-VRd quad regimens.

BOARD TRAP

WRONG answer: 'transplant-eligible patients should receive high-dose alkylator (e.g., melphalan) DURING induction.' Melphalan is AVOIDED in induction for transplant candidates because it damages the stem-cell pool and impairs later collection — it is reserved for the transplant conditioning step. Discriminator: PI+IMiD+steroid (±anti-CD38) for induction; melphalan only at conditioning.

9. Age/fitness triage

WHAT YOU SEE

Triage officer: age >70 / frailty / comorbidities

WHAT IT ENCODES

Transplant eligibility is determined by PERFORMANCE STATUS, frailty, and comorbidity burden (cardiac, pulmonary, renal, hepatic) — NOT a rigid chronologic age cutoff. Many fit patients in their early 70s undergo ASCT; conversely a frail 60-year-old may not be a candidate. Mnemonic: the triage officer weighs fitness and comorbidities, not just the birth date.

BOARD TRAP

WRONG answer: 'this 71-year-old is automatically ineligible for autologous transplant because of age.' Age ALONE does not exclude transplant — assess functional status and comorbidities. Discriminator: a fit older adult can proceed to ASCT, whereas significant frailty/organ dysfunction is the true disqualifier.

10. Maintenance

WHAT YOU SEE

Maintenance chamber, lenalidomide ongoing

WHAT IT ENCODES

After induction (and after transplant in eligible patients), continuous LENALIDOMIDE maintenance until progression improves both PFS and overall survival; high-risk patients often get a PI-containing maintenance (e.g., add bortezomib/ixazomib). Continue thromboprophylaxis and monitor counts and for second primary malignancies. Mnemonic: the maintenance chamber keeps lenalidomide running long-term.

BOARD TRAP

WRONG answer: 'maintenance therapy is given for a fixed short course after transplant.' Standard lenalidomide maintenance is CONTINUOUS until progression (or intolerance), and this duration is what drives the survival benefit. Discriminator: stopping maintenance early forfeits the PFS/OS advantage; high-risk cytogenetics warrant intensified (PI-containing) maintenance.

11. HDT + ASCT 'the bomb'

WHAT YOU SEE

High-dose melphalan bomb

WHAT IT ENCODES

Autologous stem-cell transplant = high-dose MELPHALAN (200 mg/m²) conditioning ('the bomb') to maximally cytoreduce, followed by rescue with the patient's own previously collected/cryopreserved stem cells. It deepens response (raises CR rate) and prolongs remission/PFS in eligible patients. Mnemonic: the melphalan bomb levels the marrow before the rescue.

BOARD TRAP

WRONG answer: 'autologous transplant is CURATIVE for multiple myeloma.' ASCT is NOT curative — it deepens response and extends remission but the disease invariably relapses; that is why maintenance follows. Discriminator: myeloma remains incurable with standard therapy; ASCT improves depth/duration of response, not cure.

12. Stem-cell rescue

WHAT YOU SEE

Stored autologous stem cells floating down

WHAT IT ENCODES

After melphalan conditioning, the cryopreserved AUTOLOGOUS hematopoietic stem cells are reinfused to rescue the ablated marrow and restore counts (engraftment over ~10–14 days). This rescue is what allows the otherwise marrow-lethal melphalan dose; it deepens CR by roughly 25–40%. Mnemonic: the stored stem cells parachute back down to rebuild the marrow.

BOARD TRAP

WRONG answer: 'the reinfused stem cells provide a graft-versus-myeloma immune effect.' Autologous cells are the patient's OWN — there is NO graft-versus-tumor and no GVHD; they simply restore hematopoiesis. Discriminator: graft-versus-myeloma/GVHD belong to ALLOGENEIC transplant, which is rarely used in myeloma.

13. Allogeneic rarely used

WHAT YOU SEE

Allogeneic door crossed out — high toxicity

WHAT IT ENCODES

Allogeneic transplant offers a potential graft-versus-myeloma effect but carries high treatment-related mortality and GVHD, so it is RARELY used — reserved for select young, very-high-risk or relapsed patients, ideally on a trial. Mnemonic: the allogeneic door is crossed out — 'high toxicity, rarely used.'

BOARD TRAP

WRONG answer: 'allogeneic transplant is the standard consolidative therapy for newly diagnosed myeloma.' Standard consolidation is AUTOLOGOUS transplant; allo is not standard of care due to prohibitive TRM/GVHD. Discriminator: 'autologous' = standard in eligible patients; 'allogeneic' = investigational/last-resort in myeloma.

14. Tandem/early vs delayed

WHAT YOU SEE

Timing note early vs delayed transplant

WHAT IT ENCODES

Stem cells are COLLECTED EARLY (before extended therapy compromises the pool) even if the transplant itself is DELAYED — early vs delayed ASCT show similar overall survival, so timing can be individualized. TANDEM (two sequential autologous transplants) benefits mainly HIGH-RISK cytogenetic patients, not standard-risk. Mnemonic: collect now, transplant whenever; tandem only for the high-risk crowd.

BOARD TRAP

WRONG answer: 'every myeloma patient should receive tandem (double) autologous transplant for best outcomes.' Tandem ASCT benefits only HIGH-RISK cytogenetics; for standard-risk it adds toxicity without clear survival gain. Discriminator: reserve tandem for del(17p)/t(4;14)-type high-risk disease, and always collect stem cells early regardless of transplant timing.
